EXPERIMENT -6a : Exploratory Data Analysis

AIM: To perform Exploratory Data Analysis (EDA) on the Netflix Titles dataset using Python and data visualization tools to analyze content distribution, ratings, release and addition trends, durations, regional contributions, textual descriptions, and genre co-occurrences.

ALGORITHM:

1. **Import required Python libraries including pandas, numpy, matplotlib.pyplot, seaborn, and wordcloud.**
2. **Download the netflix_titles dataset from Kaggle and load it into a Pandas DataFrame.**
3. **Explore the dataset structure by checking data types, summary statistics, missing values, and column info.**
4. **Perform Content Distribution Analysis:**
    - **Compare the counts of Movies vs. TV Shows using visualization plots.**
    - **Plot the distribution of ratings (e.g., TV-MA, PG, R) and identify countries with the highest content volume.**
5. **Conduct Time-Based Trend Analysis:**
    - **Plot release years and addition years to analyze Netflix's content growth trends over time.**
    - **Identify which months of the year Netflix adds the most content.**
6. **Conduct Duration & Content Type Analysis:**
    - **Plot the distribution of movie durations in minutes.**
    - **Plot the distribution of TV shows categorized by number of seasons.**
7. **Execute Genre & Categories Analysis:**
    - **Extract and plot the top 10 most common genres, top directors, and most frequent actors.**
    - **Analyze country-wise production of movies versus TV shows.**
    - **Determine the most common ratings specifically for Movies versus TV Shows.**
8. **Execute Text & Descriptions Analysis:**

- **Generate a word cloud on the description column to identify the most common words.**

9. **Determine the oldest and newest titles available on the platform.**
10. **Construct a genre correlation heatmap to analyze which genres frequently co-occur together.**

1. Download Netflix Titles Dataset and Convert It to DataFrame

```
#Exp_no_6a
#Sanjay_S_241801246
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from wordcloud import WordCloud
df = pd.read_csv("netflix_titles.csv")
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

2. Explore the Dataset

```
#Exp_no_6a
#Sanjay_S_241801246
print("Dataset Shape:", df.shape)
```

```
Dataset Shape: (8807, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
print(df.isnull().sum())
```

```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

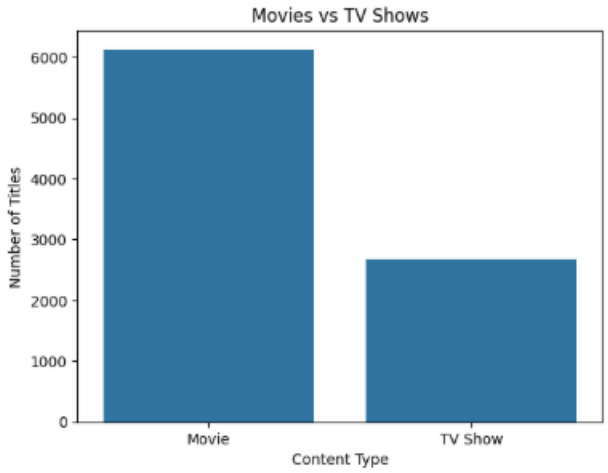
```
for col in ["director", "cast", "country", "rating", "date_added", "duration"]:
    df[col] = df[col].fillna("Unknown")

df["date_added"] = pd.to_datetime(df["date_added"], errors="coerce")
```

3a) How many Movies vs TV Shows are in the dataset?

```
#Exp_no_6a
#Sanjay_S_241801246
print(df["type"].value_counts())
sns.countplot(data=df, x="type")
plt.title("Movies vs TV Shows")
plt.xlabel("Content Type")
plt.ylabel("Number of Titles")
plt.show()
```

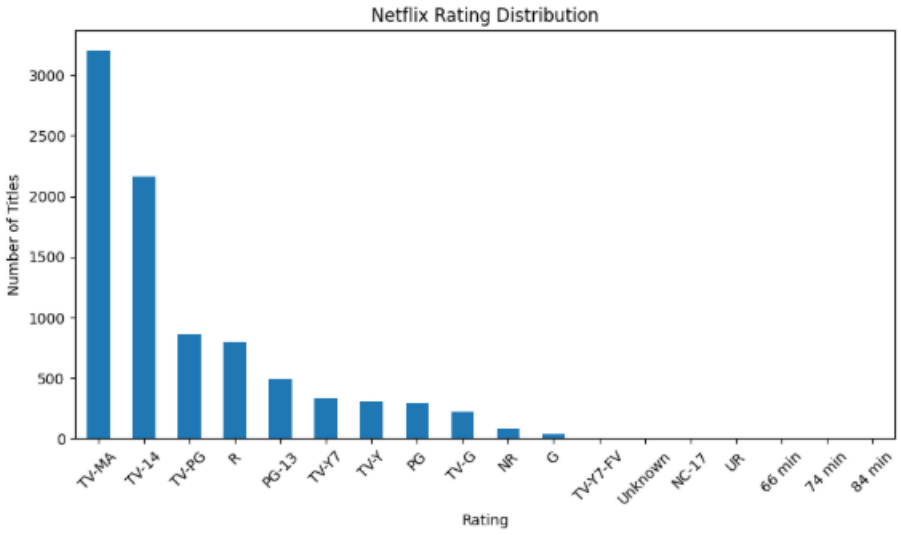
```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```



3b) What is the distribution of ratings?

```
#Exp_no_6a
#Sanjay_S_241801246
rating_counts = df["rating"].value_counts()
print(rating_counts)
rating_counts.plot(kind="bar", figsize=(10, 5))
plt.title("Netflix Rating Distribution")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

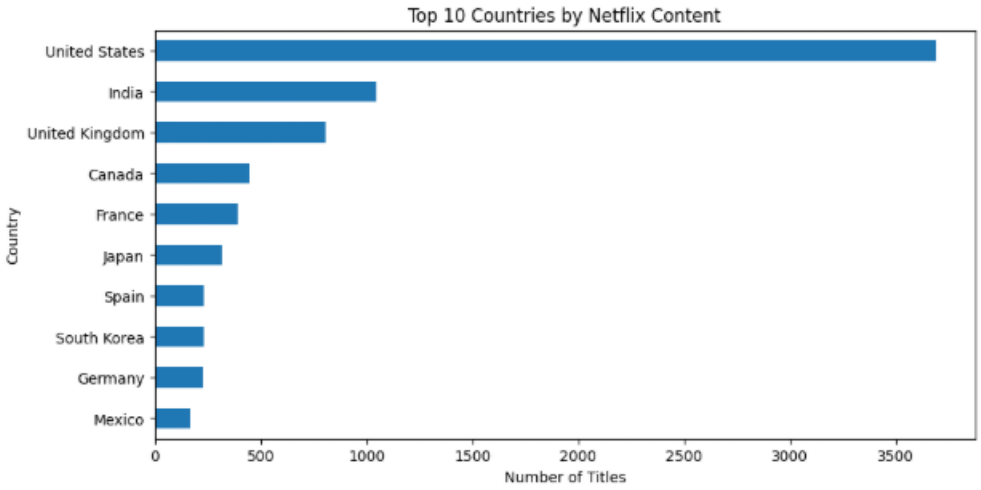
```
rating
TV-MA       3207
TV-14       2160
TV-PG        863
R            799
PG-13        490
TV-Y7        334
TV-Y         307
PG           287
TV-G         220
NR            80
G             41
TV-Y7-FV       6
Unknown        4
NC-17          3
UR             3
66 min         1
74 min         1
84 min         1
Name: count, dtype: int64
```



3c) Which countries have the most Netflix content?

```
#Exp_no_6a
#Sanjay_S_241801246
countries = df[df["country"] != "Unknown"]["country"].str.split(", ").explode()
top_countries = countries.value_counts().head(10)
print(top_countries)
top_countries.sort_values().plot(kind="barh", figsize=(10, 5))
plt.title("Top 10 Countries by Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.show()
```

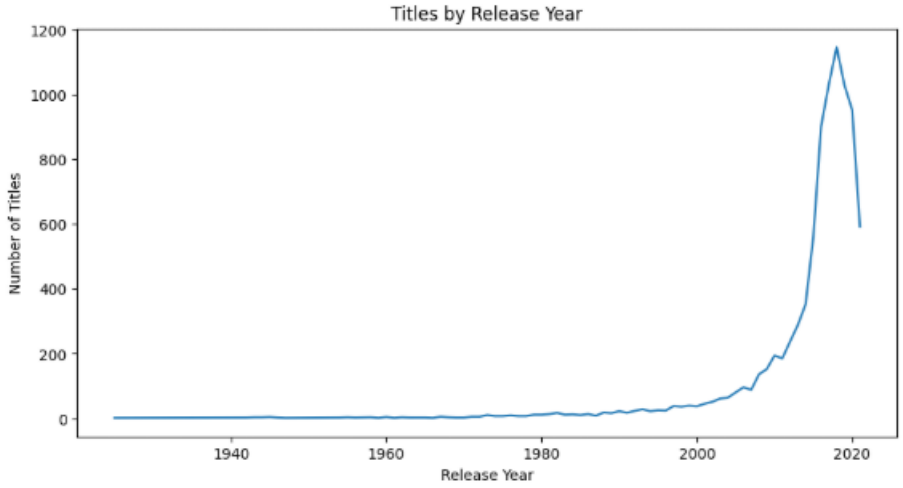
```
country
United States     3689
India             1046
United Kingdom     804
Canada             445
France             393
Japan              318
Spain              232
South Korea        231
Germany            226
Mexico             169
Name: count, dtype: int64
```



4a) In which years were the most shows/movies released?

```
#Exp_no_6a
#Sanjay_S_241801246
release_counts = df["release_year"].value_counts()
print("Top 10 Release Years:")
print(release_counts.head(10))
release_counts.sort_index().plot(figsize=(10, 5))
plt.title("Titles by Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()
```

```
Top 10 Release Years:
release_year
2018    1147
2017    1032
2019    1030
2020     953
2016     902
2021     592
2015     560
2014     352
2013     288
2012     237
Name: count, dtype: int64
```

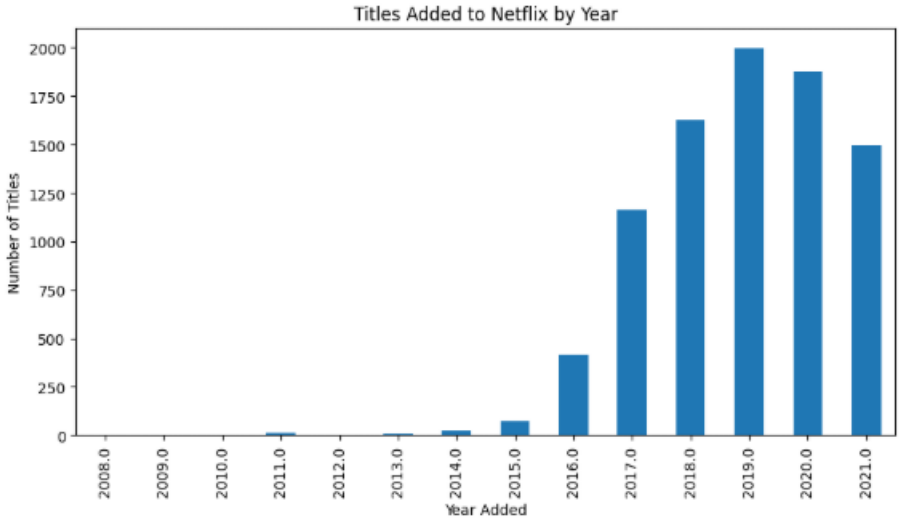


4b) In which years were titles added to Netflix?

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```
#Exp_no_6a
#Sanjay_S_241801246
df["added_year"] = df["date_added"].dt.year
added_counts = df["added_year"].value_counts().sort_index()
print(added_counts)
added_counts.plot(kind="bar", figsize=(10, 5))
plt.title("Titles Added to Netflix by Year")
plt.xlabel("Year Added")
plt.ylabel("Number of Titles")
plt.show()
```

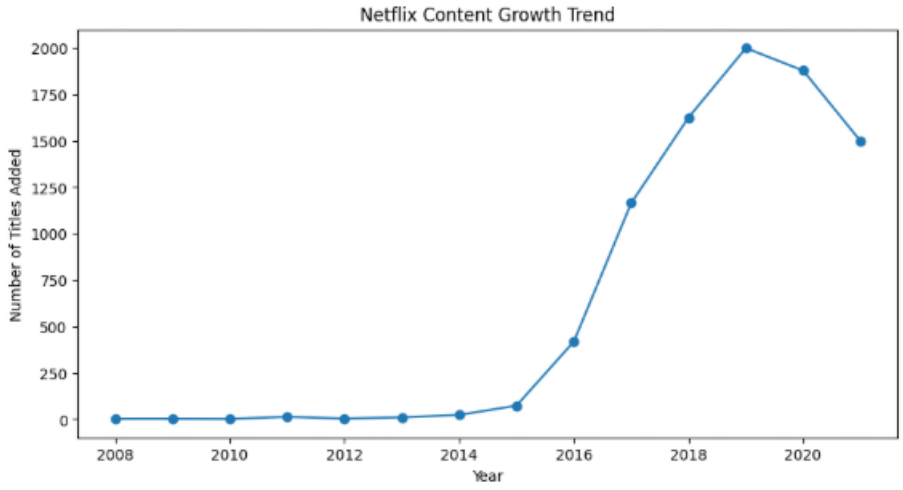
```
added_year
2008.0       2
2009.0       2
2010.0       1
2011.0      13
2012.0       3
2013.0      10
2014.0      23
2015.0      73
2016.0     418
2017.0    1164
2018.0    1625
2019.0    1999
2020.0    1878
2021.0    1498
Name: count, dtype: int64
```



4c) How has Netflix's content growth trend changed over time?

```
#Exp_no_6a
#Sanjay_S_241801246
growth = df.groupby("added_year").size().sort_index()
print(growth)
growth.plot(kind="line", marker="o", figsize=(10, 5))
plt.title("Netflix Content Growth Trend")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.show()
```

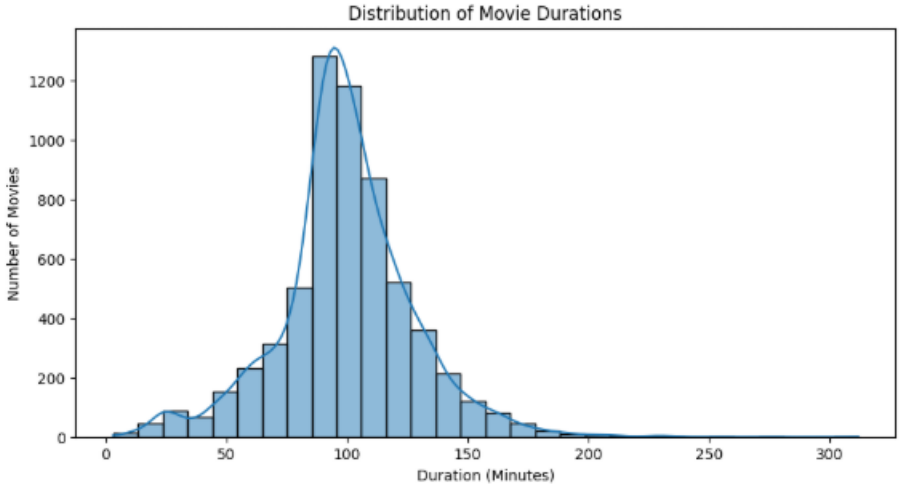
```
added_year
2008.0       2
2009.0       2
2010.0       1
2011.0      13
2012.0       3
2013.0      10
2014.0      23
2015.0      73
2016.0     418
2017.0    1164
2018.0    1625
2019.0    1999
2020.0    1878
2021.0    1498
dtype: int64
```



5a) What is the distribution of movie durations in minutes?

```
#Exp_no_6a
#Sanjay_S_241801246
movies = df[df["type"] == "Movie"].copy()
movies["duration_minutes"] = pd.to_numeric(
    movies["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)
print(movies["duration_minutes"].describe())
plt.figure(figsize=(10, 5))
sns.histplot(movies["duration_minutes"].dropna(), bins=30, kde=True)
plt.title("Distribution of Movie Durations")
plt.xlabel("Duration (Minutes)")
plt.ylabel("Number of Movies")
plt.show()
```

```
count    6128.000000
mean       99.577187
std        28.290593
min         3.000000
25%        87.000000
50%        98.000000
75%       114.000000
max       312.000000
Name: duration_minutes, dtype: float64
```

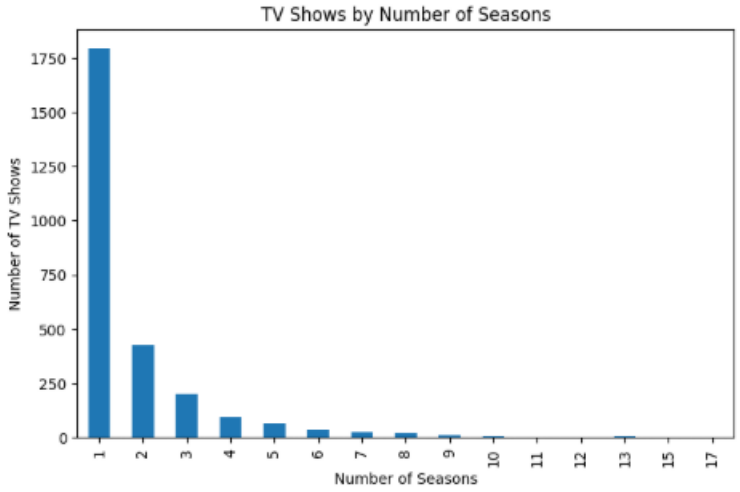


Duration (Minutes)

5b) How many TV shows exist per number of seasons?

```
#Exp_no_6a
#Sanjay_S_241801246
tv = df[df["type"] == "TV Show"].copy()
tv["seasons"] = pd.to_numeric(
    tv["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)
season_counts = tv["seasons"].value_counts().sort_index()
print(season_counts)
season_counts.plot(kind="bar", figsize=(8, 5))
plt.title("TV Shows by Number of Seasons")
plt.xlabel("Number of Seasons")
plt.ylabel("Number of TV Shows")
plt.show()
```

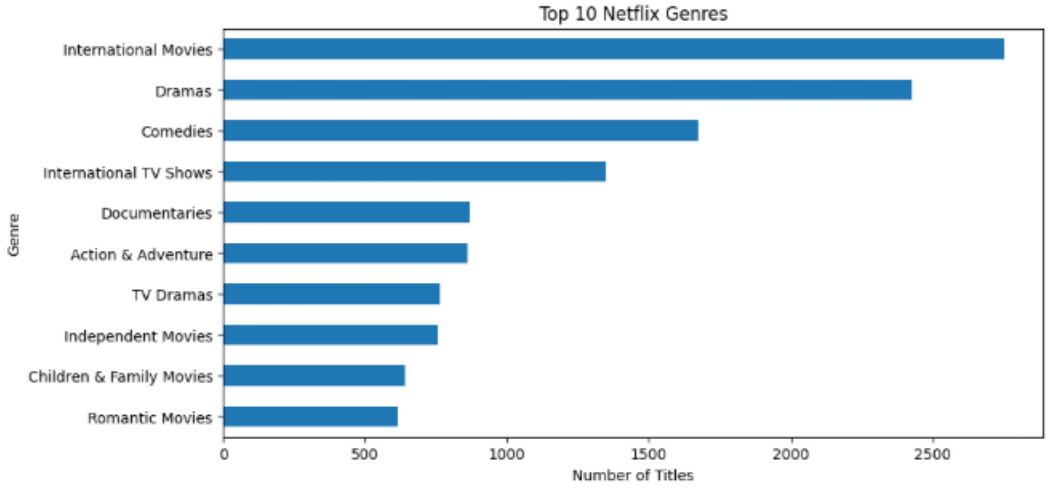
```
seasons
1     1793
2      425
3      199
4       95
5       65
6       33
7       23
8       17
9        9
10       7
11       2
12       2
13       3
15       2
17       1
Name: count, dtype: int64
```



6a) What are the top 10 most common genres on Netflix?

```
#Exp_no_6a
#Sanjay_S_241801246
genres = df["listed_in"].dropna().str.split(", ").explode()
top_genres = genres.value_counts().head(10)
print(top_genres)
top_genres.sort_values().plot(kind="barh", figsize=(10, 5))
plt.title("Top 10 Netflix Genres")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")
plt.show()
```

```
listed_in
International Movies        2752
Dramas                      2427
Comedies                    1674
International TV Shows      1351
Documentaries                869
Action & Adventure           859
TV Dramas                    763
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Name: count, dtype: int64
```

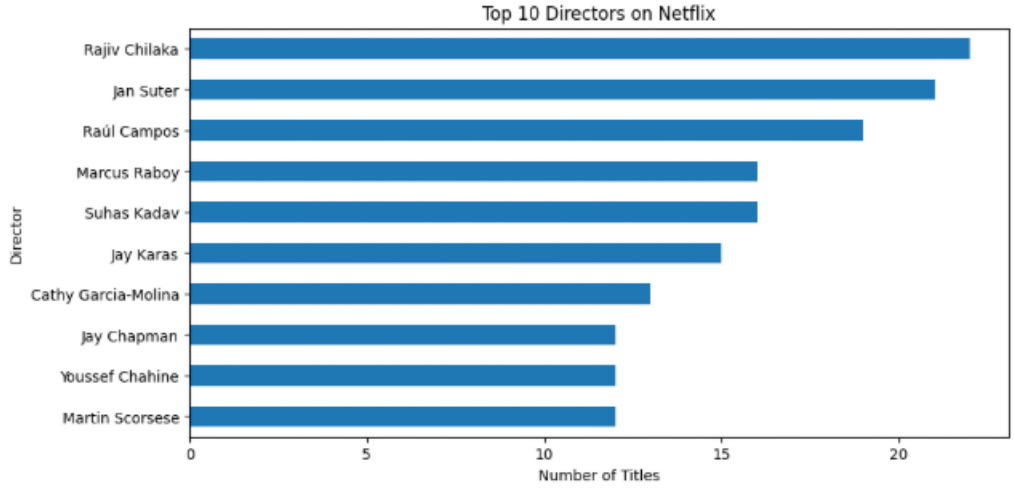


6b) Which directors have the most titles on Netflix?

6b) Which directors have the most titles on Netflix?

```
#Exp_no_6a
#Sanjay_S_241801246
directors = df[df["director"] != "Unknown"]["director"].str.split(", ").explode()
top_directors = directors.value_counts().head(10)
print(top_directors)
top_directors.sort_values().plot(kind="barh", figsize=(10, 5))
plt.title("Top 10 Directors on Netflix")
plt.xlabel("Number of Titles")
plt.ylabel("Director")
plt.show()
```

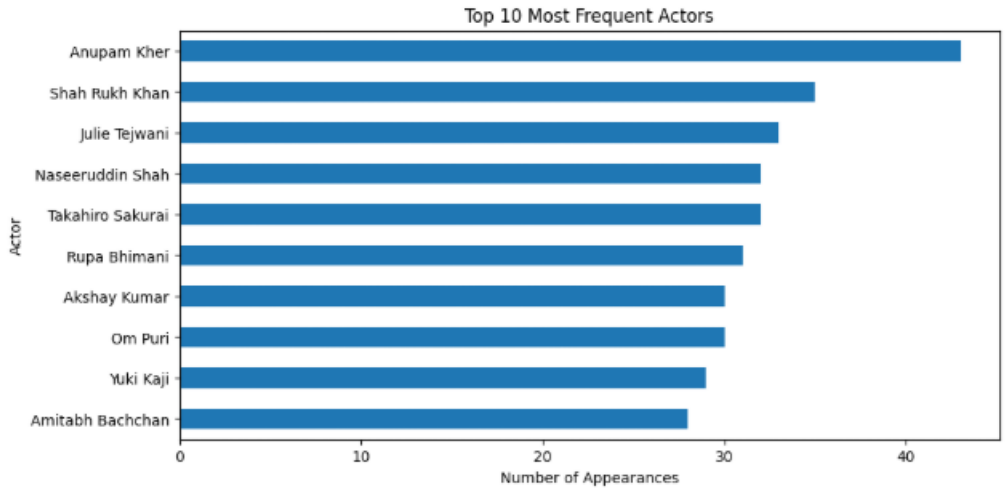
```
director
Rajiv Chilaka          22
Jan Suter              21
Raúl Campos            19
Suhas Kadav            16
Marcus Raboy           16
Jay Karas              15
Cathy Garcia-Molina    13
Martin Scorsese        12
Youssef Chahine        12
Jay Chapman            12
Name: count, dtype: int64
```



6c) Which actors appear most frequently across Netflix shows/movies?

```
#Exp_no_6a
#Sanjay_S_241801246
actors = df[df["cast"] != "Unknown"]["cast"].str.split(", ").explode()
top_actors = actors.value_counts().head(10)
print(top_actors)
top_actors.sort_values().plot(kind="barh", figsize=(10, 5))
plt.title("Top 10 Most Frequent Actors")
plt.xlabel("Number of Appearances")
plt.ylabel("Actor")
plt.show()
```

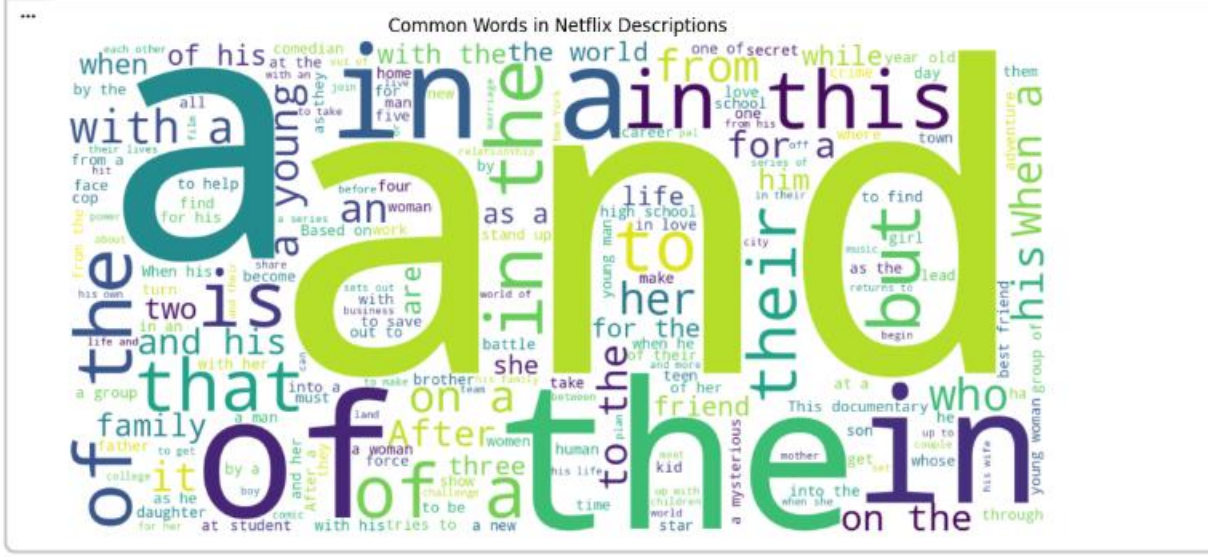
```
cast
Anupam Kher          43
Shah Rukh Khan       35
Julie Tejwani        33
Takahiro Sakurai     32
Naseeruddin Shah     32
Rupa Bhimani         31
Om Puri              30
Akshay Kumar         30
Yuki Kaji            29
Amitabh Bachchan     28
Name: count, dtype: int64
```



7. Text & Descriptions

7a) What are the most common words in the description field? (Word Cloud)

```
#Exp_no_6a
#Sanjay_S_241801246
text = " ".join(df["description"].dropna().astype(str))
wordcloud = WordCloud(
    width=1200,
    height=600,
    background_color="white",
    stopwords="english"
).generate(text)
plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation="bilinear")
plt.axis("off")
plt.title("Common Words in Netflix Descriptions")
plt.show()
```

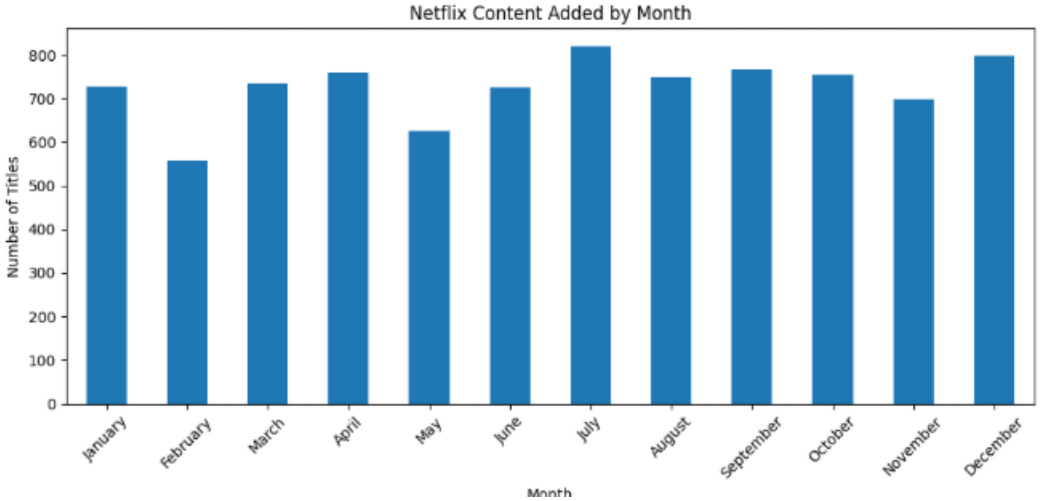


8. In Which Months of the Year Does Netflix Add the Most Content?

8. In Which Months of the Year Does Netflix Add the Most Content?

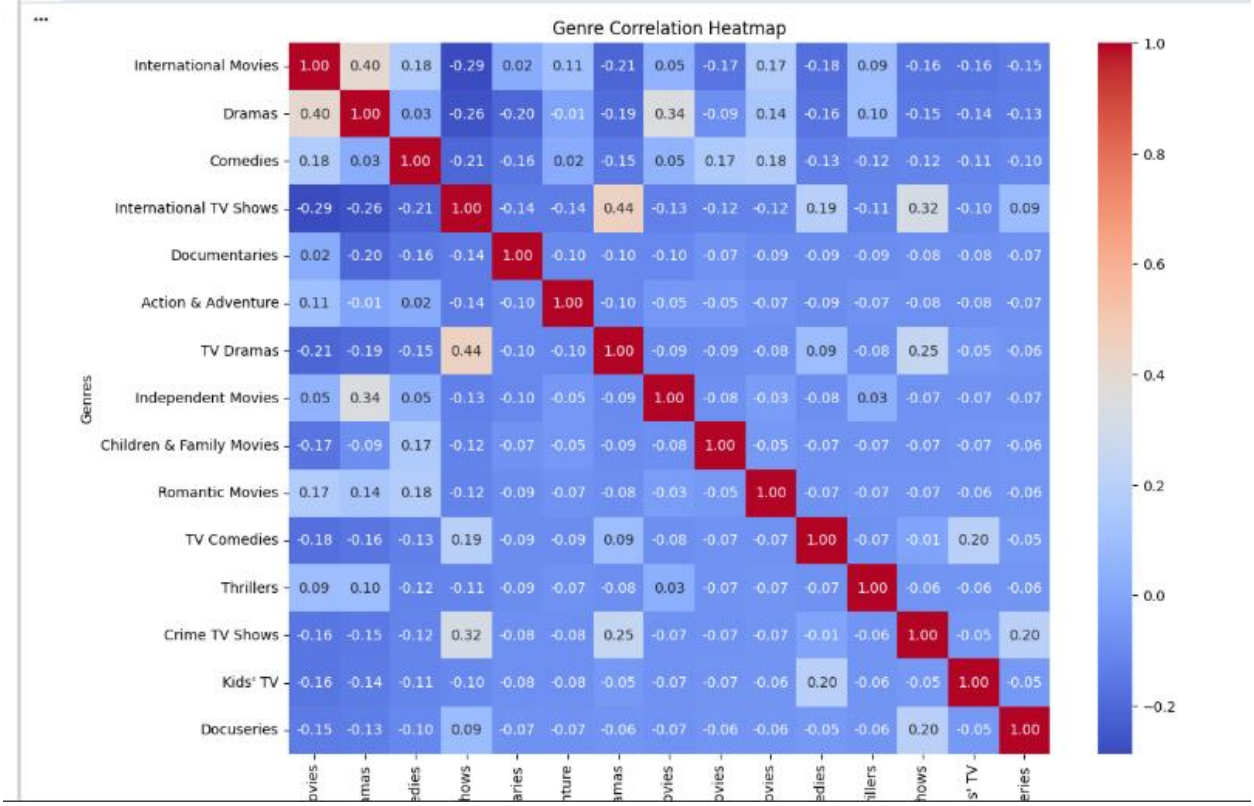
```
#Exp_no_6a
#Sanjay_S_241801246
import pandas as pd
import matplotlib.pyplot as plt
df["date_added"] = pd.to_datetime(df["date_added"], errors="coerce")
month_counts = df["date_added"].dt.month_name().value_counts()
month_order = [
    "January", "February", "March", "April",
    "May", "June", "July", "August",
    "September", "October", "November", "December"
]
month_counts = month_counts.reindex(month_order, fill_value=0)
print(month_counts)
month_counts.plot(kind="bar", figsize=(10, 5))
plt.title("Netflix Content Added by Month")
plt.xlabel("Month")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
date_added
January      727
February     557
March        734
April        759
May          626
June         724
July         819
August       749
September    765
October      755
November     697
December     797
Name: count, dtype: int64
```



12. Which Genres Frequently Co-occur Together? (Genre Correlation Heatmap)

```
#Exp_no_6a
#Sanjay_S_241801246
genre_data = df["listed_in"].fillna("").str.get_dummies(sep=", ")
top_genres = genre_data.sum().sort_values(ascending=False).head(15).index
genre_matrix = genre_data[top_genres]
correlation = genre_matrix.corr()
plt.figure(figsize=(12, 9))
sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    cmap="coolwarm"
)
plt.title("Genre Correlation Heatmap")
plt.xlabel("Genres")
plt.ylabel("Genres")
plt.show()
```



RESULT: Exploratory Data Analysis on the Netflix Titles dataset was successfully completed. Content characteristics, growth trends, duration distributions, key contributors, description keywords, and genre relationships were systematically extracted and visualized.